PV102 Day 5 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 21 | Morning Quiz - Answer Key
Storage has a defined job: carry named loads through named time. | Monday, August 10, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. Battery capacity is commonly expressed as:

- A. Ah or Wh
- B. Degrees azimuth only
- C. Wire color
- D. Roof pitch only

Answer: A

Rationale/look-for: Battery capacity may be stated in amp-hours and/or watt-hours.

2. A battery bank should be sized from:

- A. Defined loads and runtime
- B. The prettiest battery case
- C. Random parts available
- D. Module color

Answer: A

Rationale/look-for: Storage follows the job it must perform.

3. Usable capacity is often less than nameplate capacity because:

- A. Depth-of-discharge and protection limits matter
- B. Labels are decoration
- C. All loads are AC
- D. Sunlight is heavy

Answer: A

Rationale/look-for: Chemistry and protection limits affect practical usable energy.

4. Autonomy means:

- A. Time the system must carry loads without enough solar input
- B. A compass direction
- C. A breaker type
- D. A module frame dimension

Answer: A

Rationale/look-for: Autonomy is stored-energy coverage during low or no production.

5. Load shedding means:

- A. Reducing or turning off lower-priority loads
- B. Cutting random wires
- C. Ignoring the client
- D. Making voltage disappear

Answer: A

Rationale/look-for: A prioritized load plan can protect critical function.

6. Estimate: 1,000 Wh/day for 2 days = ___ Wh before losses and usable-capacity adjustment.

_____

_____

Answer: 2,000 Wh

Rationale/look-for: Daily energy x autonomy days.

7. Why does battery voltage matter in system design?

_____

_____

Answer: Power/current/equipment match

Rationale/look-for: Higher system voltage reduces current for the same power and must match equipment.

8. Name one battery safety boundary.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept OCPD, correct conductor size, terminals covered, BMS limits, ventilation as needed, temperature, no short circuits, manufacturer instructions.

9. Why is “more battery” not always the best answer?

_____

_____

Answer: Cost/risk/charge ability

Rationale/look-for: More storage costs more, occupies space, adds weight and protection needs, and must be rechargeable by the array.

10. Write the Day 5 rule in your own words.

_____

_____

Answer: Defined storage job

Rationale/look-for: Look for named loads and named runtime.